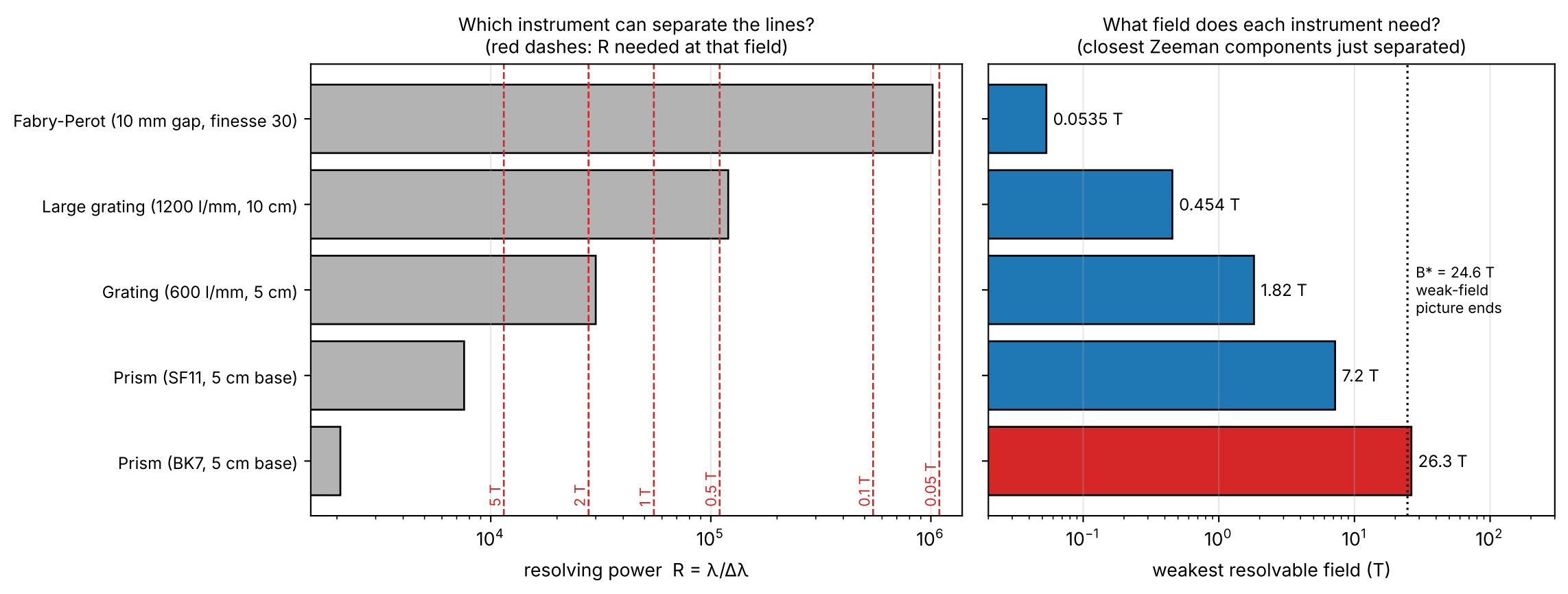
Which instrument can separate the lines?
(red dashes: R needed at that field)
Fabry-Perot (10 mm gap, finesse 30)
Large grating (1200 l/mm, 10 cm)
Grating (600 l/mm, 5 cm)
Prism (SF11, 5 cm base)
Prism (BK7, 5 cm base)
5 T
2 T
1 T
0.5 T
0.1 T
0.05 T
10^4
10^5
10^6
resolving power R = λ/Δλ
What field does each instrument need?
(closest Zeeman components just separated)
0.0535 T
0.454 T
1.82 T
7.2 T
26.3 T
B* = 24.6 T
weak-field
picture ends
10^{-1}
10^0
10^1
10^2
weakest resolvable field (T)